Traffic Signal Control

I Definitions

1. **Traffic movement**: (l, m) the traffic travelling across an intersection from one entering lane to an exiting lane;
2. **Signal phase**: Eight phases in City Brain;
3. **Pressure of each signal phase** s: $\sum_{(l,m)} x(l, m), \forall (l, m) \in s$, where (l, m) is among the permissible traffic movements;
4. **Pressure of an interaction**: $\sum$#vehicles in entering lanes - $\sum$#vehicles in exiting lanes.

II Methods

i. FRAP

1. state: the number of vehicles f_i^v on each traffic movement i and current traffic signal phase (represented as one bit f_i^s for each traffic movement signal).
2. Reward: The average queue length of each traffic movement.

Main focus: Algorithmic Framework. They adopted the distributed framework Ape-X DQN, in which the standard DRL is decomposed into *acting* and *learning*. **Acting**: assigns multiple actors with different exploration policies to interact with an environment and to store the observed date in a replay memory; **Learning**: sampling the training data in the replay memory to update the model.

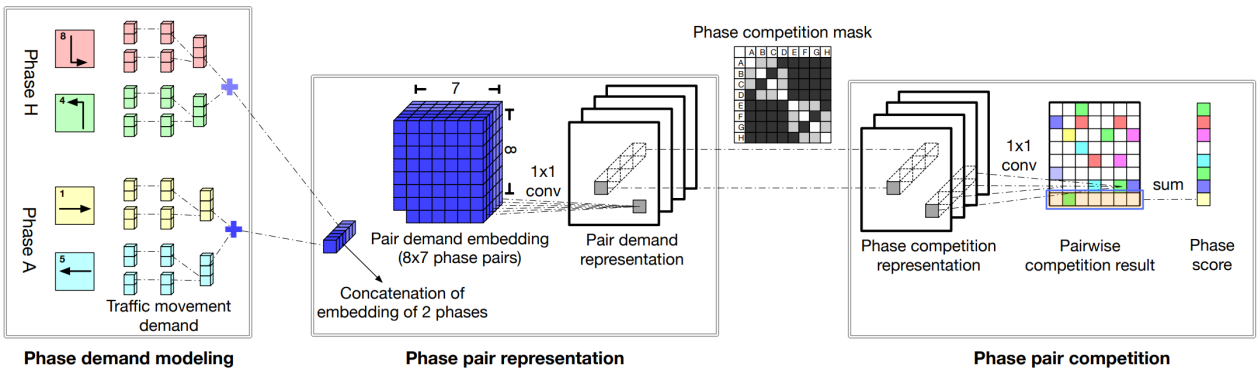

图 1: Network Design of FRAP

The prediction of Q-values has been divided into three steps:

1. *Phase Demand Modeling.* Input: number of vehicles, current phase; Output: demand for green signal. $(h_i^v = ReLU(W^v f_i^v + b^v), h_i^s = ReLU(W^s f_i^s + b^s), d_i = ReLU(W^h [h_i^v, h_i^s] + b^h))$
2. *Phase Pair Embedding* Construct $\mathbf{D}$, and then apply K convolutional layers with 1×1 filters on $\mathbf{D}$.
3. *Phase Pair Competition.* $\mathbf{H}_k^r = ReLU(W_k^r \cdot \mathbf{H}_{k-1}^r + b_k^r), \mathbf{H}_k^d = ReLU(W_k^d \cdot \mathbf{H}_{k-1}^v + b_k^d)$, where $H_0^r = \mathbf{E}$ and $\mathbf{H}_0^d = \mathbf{D}$.

After all these, we have $C = ReLU(W^c \cdot \mathbf{H}^c + b^c)$.

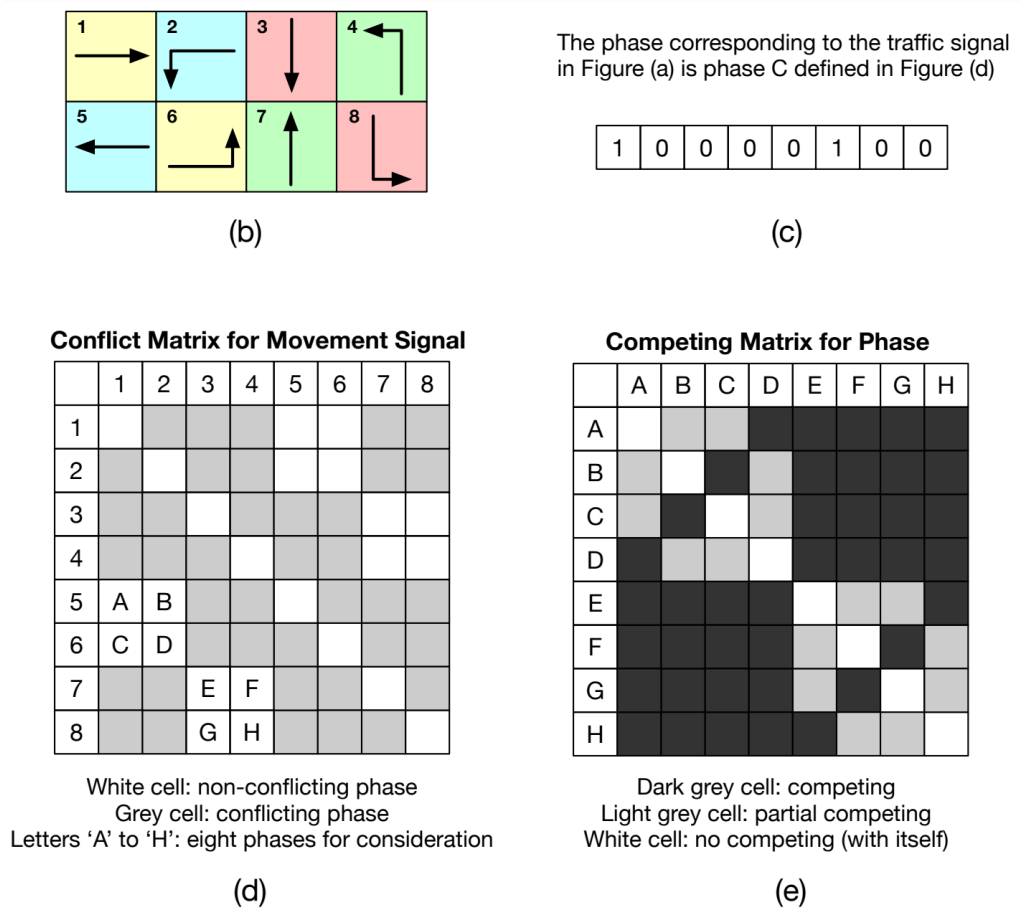

图 2: Illustration of preliminary definition in FRAP

ii. CoLight

Contribution: (1) Cooperation through dynamic communication. (Use graph attentional network to learn the dynamics of the influences.) (2) Index-free model learning with parameter sharing. (Averaging over the influences of all neighboring intersections with the learned attention weights, instead of using a fixed indexing for neighbors.)

1. Observation: $o \in \mathcal{O}$: current phase and the number of vehicles on each lane connected with the intersection

Main focus: (1) Observation modification: Graph Attention Networks for Cooperation. (2) Q-value Prediction.

1. Observation embedding: $h_i = ReLU(o_i W_e + b_e)$;
   (1) *Observation Interaction*: $e_{ij} = (h_i W_t) \cdot (h_j W_s)^T$.
   (2) *Attention Distribution within Neighborhood Scope.* $\alpha_{ij} = softmax(e_{ij}) = \frac{\exp(e_{ij}/\tau)}{\sum_{j \in \mathcal{N}_i} \exp(e_{ij}/\tau)}$.
   (3) *Index-free Neighborhood Cooperation.* $hs_i = \sigma(W_q \cdot \sum_{j \in \mathcal{N}_i} \alpha_{ij}(h_j W_c) + b_q)$.
   (4) *Multi-head Attention.*
2. Q-value Prediction:

$$hi = Embed(o_i^t); \quad hm_i^1 = GAT^1(h_i); \quad hm_i^L = GAT^L(hm_i^{L-1}); \quad \tilde{q}(o_i^t) = hm_i^L W_p + b_p$$

Then the loss function for CoLight to optimize the current policy is:

$$L(\theta) = \frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{N}(q(o_i^t, a_i^t) - \tilde{q}(o_i^t; a_i^t, \theta)^2) \tag{1}$$

iii. PressLight

The pressure of movement (l, m) is defined as: $w(l, m) = \frac{x(l)}{x_{max}(l)} - \frac{x(m)}{x_{max}(m)}$; where $x_{max}(l)$ is the maximum permissible vehicle number on l.

1. State(Observation). current phase p, #vehicles on each outgoing lane $x(m), (m \in L_{out})$, #vehicles on each segment of every incoming lane $x(l)_k (l \in L_{in}, k = 1, \cdots, K)$.
2. Reward. $r_i = -P_i$, where P_i is the pressure of intersection i.

iv. MPLight

Base model: FRAP; state and reward from PressLight.

1. Obsersvation. Current phase p and the pressure of 12 traffic movements.
2. Reward. $r_i = -P_i$, where P_i is the pressure of intersection i.